Statement of Purpose

Raman Kumar Jha (CE PhD applicant Fall 2026)

My research journey began in computer vision, focused on the complex problem of how machines see. As a researcher at the **Indian Institute of Technology Madras(IIT Madras)**, I developed cross-attention networks to enhance images in low-light, and as a Machine Learning Intern, I built and deployed generative AI pipelines for face dubbing. I excelled at building models to interpret the world, but I grew increasingly fascinated by a more significant challenge: How can an agent use this understanding to act within it? This desire to bridge the gap from passive perception to embodied action is what drove me to robotics. For robots to become truly autonomous, they must be able to perceive, learn, and act reliably in the world's most challenging and unstructured environments. My research aims to build these foundational capabilities, from robust perception to learning in adverse conditions for autonomous systems. With these, I wish to develop algorithms that can provide autonomy to robots, so that they can perform real-world tasks and adapt to new conditions. To build this rigorous foundation, I deliberately curated my master's curriculum at **New York University(NYU)** to master the complete robotics stack, from foundational courses in robotics to advanced courses in perception, reinforcement learning, and autonomous space robots. This transition required immense effort, but it solidified my commitment to becoming a researcher who can build, test, and deploy full-stack autonomous systems that perceive, learn, and act reliably in the real world.

My journey into robotics began with the fundamental problem of perception. At the **Computational Imaging Lab, IIT Madras**, under **Prof. Kaushik Mitra** and **Prof. Aswin Sankarnarayanan** from **Image Science Lab, Carnegie Mellon University**, I tackled the challenge of vision in low-light environments. The networks with only RGB inputs performed poorly on the extremely low-light images, due to the missing information in the RGB images; however, thermal images can capture these fundamental features, which include textures and edges. So, I proposed a new cross-attention network that leverages RGB and thermal images to enhance low-light images in dark conditions, surpassing previous state-of-the-art networks on LLVIP, and my own proposed real-world **V-TIEE dataset**. I also collected an aligned RGB-Thermal dataset for HDR image generation and low-light image enhancement, which I have released with the oral paper at the **International Conference on Image Processing(ICIP) 2025** [**1**]. This project was targeted for autonomous driving and assisted systems, and it grounded my expertise in computer vision and taught me the value of multi-modal sensing to overcome perceptual ambiguity. My expertise in the subject was recognized when I was appointed as a **Teaching Assistant(TA)** for the **Modern Computer Vision** course, where I created deep learning tutorials and guided students through their projects.

Seeking to apply these perception skills to mobile agents, I joined the **Agile Robotics and Perception Lab** at **NYU** under **Prof. Giuseppe Loianno**. Here, I moved from understanding 2D images to the more complex problem of building scalable 3D world models. My work was on adaptive keyframe selection for scene reconstruction in dynamic and unstructured environments. The proposed module selects the keyframes based on photometric error and SSIM error, which are also dynamically adapted based on the dynamics and motion of the scene. In this way, it reduced the count of keyframes used by the previous temporal keyframe selection method, alongside enhancing the reconstruction quality on all of the metrics on 6 static, and dynamic datasets. This work led to a paper under review at the **International Conference on Robotics, Computer Vision and Intelligent Systems(ROBOVIS) 2026** [**2**]. This work was pivotal, as it taught me how to create consistent spatial-temporal representations of the world that are essential for navigation. I also gained the deployment experience of the system on the drone, as well as tested it in real-world scenarios.

I believe a robot must not only see the world but act within it. To that end, I additionally

developed my skills in planning and control. I have worked on the **Virtually Integrated Project(ViP)** at **NYU** named **Self Drive** under **Prof. Chen Feng**, where I developed an **autonomous driving system** for real-world navigation within a maze. I designed and integrated perception, planning, and control modules for robust decision-making. At the final competition, my system was the **runner-up** for this self-drive ViP competition. Besides this, I have also worked on a **Diffusion-based model predictive control(MPC)** to improve robotic trajectory optimization in uncertain and dynamic environments. I combined diffusion models with MPC and implemented probabilistic trajectory generation using diffusion, which consequently increased the trajectory optimization performance over traditional MPC approaches. These projects solidified my understanding of the complete robotics pipeline and ignited my interest in how rich perceptual systems can inform more intelligent and robust decision-making for autonomous systems. My faculty at **NYU** also recognized this expertise, appointing me to **TA** roles for **Advanced Topics in Computer Vision** and **Physics-Informed Machine Learning**.

My Master's thesis at the **Embodied AI and Robotics Lab (AIR)** at **NYU**, under the supervision of **Prof. Yi Fang**, addresses the critical challenge of **Object Goal Navigation in degraded, low-light environments**. While initially experimenting with the lab's zero-shot GAMap framework, I identified that mapping-based approaches struggle with latency and robustness in low-SNR conditions. To overcome this, I pivoted to an end-to-end learning approach, successfully implementing the state-of-the-art PIRLNav framework. However, standard PIRLNav fails in the dark due to **semantic blindness** as RGB features vanish. To solve this, I am actively creating a novel "Dark-Habitat" benchmark by systematically degrading the HM3D dataset and re-designing the PIRLNav architecture to integrate multi-modal depth fusion, hypothesizing that geometric features can compensate for RGB failure. Initial results show significant recovery in navigation success, and we are targeting a submission to RA-L.

I am specifically seeking to pursue my PhD at **Texas A&M University** because the Department of **Electrical and Computer Engineering** fosters exactly the kind of interdisciplinary environment required to bridge the **"perception-action"** gap for real-world systems. I am drawn to the department's unique emphasis on **field robotics and resilient autonomy**, which moves beyond lab-controlled experiments to build robust agents capable of operating in unstructured, outdoor environments. Access to facilities like the **Center for Autonomous Vehicles and Sensor Systems (CANVASS)** would allow me to validate my algorithms on diverse platforms, ensuring my work translates from simulation to the challenging outdoor conditions.

In terms of research alignment, I see strong synergies with several faculty members. I am particularly interested in working with **Prof. Varun Murali.** My prior work on adaptive keyframe selection for scalable **3D reconstruction and Self Drive ViP** aligns directly with his interest in **SLAM and planning for outdoor environments**, and I am eager to explore how we can integrate **LLMs to give field robots high-level reasoning capabilities in unstructured settings**. Additionally, **Prof. Zhiwen Fan's** work on **world modeling from multimodal physical data** offers a compelling opportunity to extend my work on **3D reconstruction in dynamic environments**. Finally, given my interest in **learning-based control**, I would be excited to collaborate with **Prof. Dileep Kalathil** to develop **reinforcement learning algorithms** that are **sample-efficient and robust enough for deployment on physical systems**, building on my recent work in Diffusion-based MPC.

My long-term goal is to pursue a career in **academia**, where I can lead a research group dedicated to the next generation of **autonomous systems**. I am confident that the rigorous technical training and collaborative culture at **Texas A&M University** is the ideal environment to shape me into an impactful researcher and educator. I am fully prepared to contribute to the ECE department's diverse research community and look forward to the opportunity in embodied intelligence.

References

[1] Raman Jha, Adithya Lenka, Mani Ramanagopal, Aswin Sankaranarayanan, and Kaushik Mitra. Rt-x net: Rgb-thermal cross attention network for low-light image enhancement. In *2025 IEEE International Conference on Image Processing (ICIP)*, pages 1492–1497, 2025.

[2] Raman Jha, Yang Zhou, and Giuseppe Loianno. Adaptive keyframe selection for scalable 3d scene reconstruction in dynamic environments. *arXiv preprint arXiv:2510.23928*, 2025.